TP Etherchannel (LACP et PAgP)

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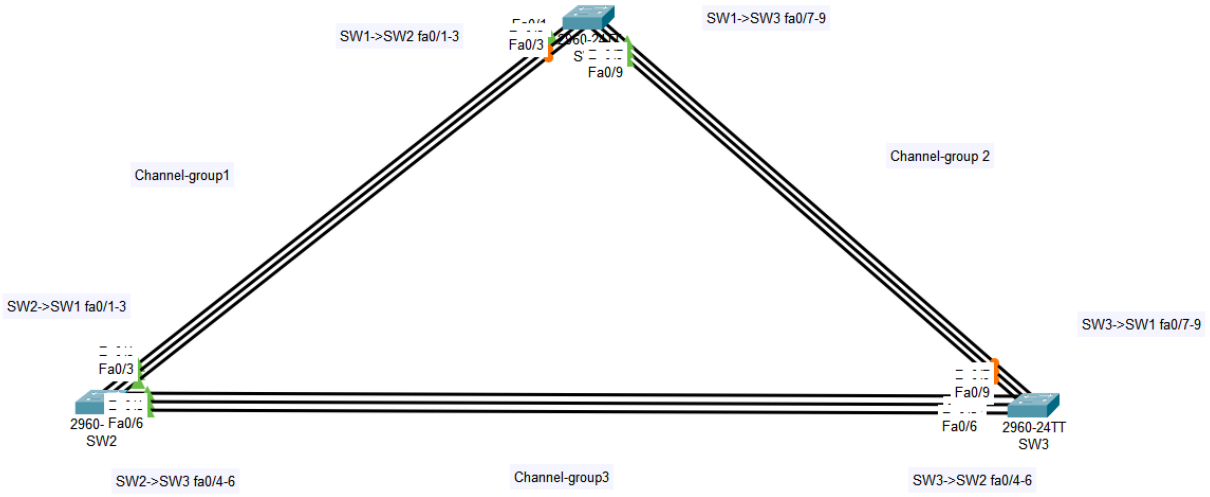
Topologie du réseau



Configuration de SW1

Ce commutateur prendra l'initiative (mode Desirable) sur les 2 channel-group qui le concerne

```
SW1> enable
SW1# configure terminal
SW1(config)# interface range fa0/1-3
SW1(config-if-range)# channel-group 1 mode desirable
SW1(config-if-range)# interface range fa0/7-9
SW1(config-if-range)# channel-group 2 mode desirable
```

```
SW1>en
SW1#config t
Enter configuration commands, one per line.  End with CNTL/Z.
SW1(config)#int
SW1(config)#interface range fa0/1-3
SW1(config-if-range)#chann
SW1(config-if-range)#channel-group 1 mode desirable
SW1(config-if-range)#
Creating a port-channel interface Port-channel 1

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/3, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/3, changed state to up

SW1(config-if-range)#interface range fa0/7-9
SW1(config-if-range)#channel-group 2 mode desirab
SW1(config-if-range)#channel-group 2 mode desirable
SW1(config-if-range)#
Creating a port-channel interface Port-channel 2

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/7, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/7, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/8, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/8, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/9, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/9, changed state to up
```

Configuration de SW2

Ce commutateur ne prendra pas l'initiative (mode Auto) que sur le channel-group 1

```
SW2> enable
SW2# configure terminal
SW2(config)# interface range fa0/1-3
SW2(config-if-range)# channel-group 1 mode auto
SW2(config-if-range)# interface range fa0/4-6
SW2(config-if-range)# channel-group 3 mode desirable
```

```
SW2>en
SW2#config t
Enter configuration commands, one per line.  End with CNTL/Z.
SW2(config)#inter
SW2(config)#interface rang
SW2(config)#interface range fa0/1-3
SW2(config-if-range)#channel-group 1 mode auto
SW2(config-if-range)#
Creating a port-channel interface Port-channel 1

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/3, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/3, changed state to up

SW2(config-if-range)#inter
SW2(config-if-range)#interface range fa0/4-6
SW2(config-if-range)#channel-group
%LINK-5-CHANGED: Interface Port-channel1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Port-channel1, changed
SW2(config-if-range)#interface range fa0/4-6
SW2(config-if-range)#channel-group 3 mode desirable
SW2(config-if-range)#
Creating a port-channel interface Port-channel 3

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/4, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/4, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/5, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/5, changed state to up
```

Configuration de SW3

Ce commutateur ne prendra pas l'initiative (mode Auto) sur les 2 channel-group qui le concerne

```
SW3> enable
SW3# configure terminal
SW3(config)# interface range fa0/4-6
SW3(config-if-range)# channel-group 3 mode auto
SW3(config-if-range)# interface range fa0/7-9
SW3(config-if-range)# channel-group 2 mode auto
```

```
SW3#config t
Enter configuration commands, one per line.  End with CNTL/Z.
SW3(config)#inter
SW3(config)#interface range fa0/4-6
SW3(config-if-range)#channel-group 3 mode auto
SW3(config-if-range)#
Creating a port-channel interface Port-channel 3

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/4, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/4, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/5, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/5, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/6, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/6, changed state to up

SW3(config-if-range)#interface range fa0/7-9
SW3(config-if-range)#channel
%LINK-5-CHANGED: Interface Port-channel3, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Port-channel3, changed statechannel
% Ambiguous command: "channel"
SW3(config-if-range)#channel-group 2 mode auto
SW3(config-if-range)#
Creating a port-channel interface Port-channel 2

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/7, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/7, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/8, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/8, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/9, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/9, changed state to up
```

Vérification sur SW1

```
SW1 (config)# do show etherchannel summary
```

```
SW1(config)#do show etherchannel summary
Flags:  D - down        P - in port-channel
        I - stand-alone s - suspended
        H - Hot-standby (LACP only)
        R - Layer3      S - Layer2
        U - in use      f - failed to allocate aggregator
        u - unsuitable for bundling
        w - waiting to be aggregated
        d - default port


Number of channel-groups in use: 2
Number of aggregators:           2

Group  Port-channel  Protocol    Ports
------+-------------+-----------+-----------------------------------------------

1      Po1(SU)           PAgP   Fa0/1(P) Fa0/2(P) Fa0/3(P)
2      Po2(SU)           PAgP   Fa0/7(P) Fa0/8(P) Fa0/9(P)
```

Le flag SU montre que l'agrégation est active et fonctionnelle

Commande pour ajouté de la sécurité à une interface agrégée

```
SW1(config)# interface port-channel 1
```